



Buying Tips

- ✓ **INPUT VOLTAGE RANGE:** This is the range beyond which the UPS delivers power through its batteries. The greater the range the UPS can handle, the more versatile its power protection. Look for a voltage range between 150 and 270 V.
- ✓ **VA RATING:** This is the power rating that can be provided by the UPS. Choose the rating based on the requirements of the equipment you intend using with the UPS. As a rule of thumb, a 1 GHz processor-based system with a 20 GB hard disk, a CD-ROM drive and a 17-inch colour monitor would require a 500 VA UPS.
- ✓ **BACKUP TIME:** This is the specified time for which the UPS can provide power from its battery for a given load. Look for a backup time of at least 10 minutes at maximum load.
- ✓ **SWITCHING OR TRANSFER TIME:** This is the time taken by the UPS to switch from normal delivery power mode to battery mode on detecting a power failure. Ideally, look for transfer times under 5 milliseconds.
- ✓ **ALARMS AND CONNECTIVITY:** Most UPS systems have front panel LEDs that indicate the battery level and the condition of the input and output power. This comes in useful, especially during power failures.
- ✓ **CHARGING TIME:** This is the time taken to charge the UPS batteries from the completely drained state to the fully charged condition. The charging time should be between 4 and 6 hours.

- 1 **Battery:** The heart of a UPS; the full-load capability of a UPS is directly dependent on the battery.
- 2 **LED and power button:** Pressing the power button starts off power at the AC-out connectors. There are LEDs to indicate power, standby, and overload conditions.
- 3 **Voltmeter:** Indicates the current voltage in as well as voltage out.
- 4 **AC out:** Power connections to the cabinet SMPS, the display and the speakers come from here.

Quick Find

YOU NEED	To provide power to a low or medium configuration consumer PC
Basic Use	
LOOK FOR	An offline or line-interactive UPS, with a 450 VA to 650 VA rating, and a backup time of at least 10 minutes
YOU NEED	To provide power to a PC using high-end components or a server, so as to have minimum downtime
Inter Use	
LOOK FOR	An online UPS with at least 1 KVA rating, and a backup time of at least 10 minutes. It should also support remote monitoring via a serial or USB interface, with alarms and indicators to warn of power anomalies

Do Remember

- Decide on the UPS based on the total power consumed by your equipment. If it's expressed in Amps, then multiply that value by 230 V—the nominal line voltage. If the power drawn is stated in Watts, multiply it by a factor of 1.3 to 1.4 for the VA load. For example, a 250 Watt power supply (x 1.4) equals a 350 VA load. Considering a measure of safety, 500 VA is enough for most PCs.
- The VA rating of a UPS is a measure of the amount of power it can provide—usually for about 10 to 12 minutes. Therefore, if a 500 VA UPS is loaded with equipment that draws 500 VA of power, it will last for about 10 minutes. If the load is halved to 250 VA, the backup time will increase to about 16 to 18 minutes. Of course, this is assuming that the battery is fully charged and functional. Always ascertain that the equipment is connected to the UPS.
- It is advisable to charge and discharge the UPS battery completely once a month to prolong battery life. To do this, run the computer from the battery until it is completely discharged, and then recharge it.



The Future

The next trend in the Indian UPS industry should see a major shift towards reliable and intelligent products. UPSes are as important for mission-critical applications as any other element, and reliability is a major issue. At the same time, intelligent products will be able to maximise power availability for the equipment that needs it most, shutting down non-necessary equipment at different stages of battery usage. Running costs will also be a driving factor.